

Simulation study design for investigating Random Forests for Variable Importance

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1 Purpose of simulation study

The purpose of the present simulation study is to investigate the performance of various variable importance measures for random forests. An intermediate goal is to find a parameter set for random forest fitting that yield good results in a situation similar to Hindborgs study of ECG abnormalities in relation to OHCA.

Two situations that appear in practice that we would like a method to handle well are

1. strong correlations between a large group of variables where only one among them has a causal effect and
2. strong interaction effects between variables that individually have modest or no effect.

2 Simulation setting

2.1 Data generating process

We will simulate from a logistic regression model with a single binary outcome, Y and 52 binary covariates, $(X_1, \dots, X_{52} = X)$.

$$EY = \text{logis}(\theta_0 + \theta X) \tag{1}$$

The simulation engine is created from the ECG + OHCA dataset using the `synthesize` function of the `riskRegression` package. For this reason, the correlation structure of the covariates and the effect sizes are similar to this case.

To examine performance under different effect sizes/strength of signal, we will simulate data from different models where θ is multiplied by a parameter λ .

2.2 Adding interaction effects

2.3 Setting with no first-order effects but presence of interaction effects

2.4 Hyperparameters

Initially, we will work with a sample size of 2000. We will investigate performance of each variable importance measure under consideration for three values for each of the following parameters for random forest fitting: `mtry`, `nodesize`, `ntree`. We will also investigate performance for different scales of effect size in the data generating process.

Concretely, we will use the following values:

Parameter	Low	Medium	High
<code>mtry</code>	3	8	16
<code>nodesize</code>	1	5	20
<code>ntree</code>	250	500	2000
λ	0.1	0.55	1

Table 1: Random forest parameter values considered in the simulation study.

3 Variable importance measures

For random forests we will consider permutation-based variable importance as well as minimal depth and average treatment effect as variable importance measures. Random forests will be fit using Brier score as loss function. For a comparison outside the random forest framework, we will use average treatment effect within a logistic regression model.

4 Performance assessment

4.1 Signal strength

We ask ourselves the question: Do the variables in our model actually predict the outcome in question? To assess this, we will, for each model, compare the prediction error of the model to a trivial model that uniformly assigns the marginal average outcome of the entire population to each individual.

4.2 Notions of scoring for rankings

Exactly how performance of variable importance measures is to be scored is unclear to me at this time. A non-parametric method of scoring would be preferable, as different notions of importance might not be directly comparable. If each variable importance measure admits a ranking (preferably allowing for

ties), the accuracy of importance rankings may be evaluated against a ground truth of importance rank. The task of selecting a subset of important variables from a large set of mostly unimportant variables may be different than ranking the importance of a relatively small set of variables that is known to have some importance. These two tasks may be in tension for some methods. Perhaps it should be further clarified what kind of task we wish to score with respect to.

4.3 Ranking variable importance in the presence of interactions

When the data generating mechanism is a logistic regression with no interaction terms, the size of the parameters of the model admit a straightforward ground truth of variable importance. It is unclear to me how a one-dimensional ground truth of variable importance is to be constructed in the presence of interactions. Perhaps a computationally efficient version of Shapley values can be implemented for random forests?